

EXECUTIVE BRIEF & METHODOLOGY

Airline Passenger Satisfaction & Retention

A Practical Guide for Management: Key Findings, Surprises & Action Plan

Target: Executive Leadership

Data: 103,904 Passenger Records

Focus: Non-Technical Summary

1. THE BUSINESS PROBLEM & CORE QUESTIONS

THE PROBLEM: HIGH DISSATISFACTION

56.1% of our flyers are neutral or unhappy. In today's market, passengers can switch to another airline in 30 seconds. Losing repeat travelers hurts predictable revenue and increases customer support costs.

THE SOLUTION: SMART INVESTMENTS

Overhauling aircraft is expensive. Instead of guessing, this analysis identifies the **exact service touchpoints** that create happy flyers, so we only spend money where it actually moves the needle.

Questions We Answered: (1) Which services make or break a flight? (2) Why are Economy flyers so unhappy compared to Business Class? (3) Do flight delays automatically ruin the relationship? (4) Can a good mobile app make up for flight issues?

2. WHAT WE DID WITH THE DATA (IN PLAIN ENGLISH)

We analyzed real-world feedback from over **103,000 travelers** across 14 service areas, demographics, and flight delays. Before analyzing, we cleaned the raw data to ensure accurate results:

What Was Cleaned	Why It Matters	Result for Management
Missing Delay Records	83 flights had departure delays logged but arrival delays blank.	Filled using departure times (\$r = 0.96\$). Avoided dropping real customers.
Messy Category Names	Customer types had mixed uppercase and lowercase text.	Standardized so reporting filters don't create duplicate broken segments.
Extreme Delays (Outliers)	Delays ranged from 0 minutes to 18+ hours.	Grouped into tiers (On-Time, Minor, Moderate, Major) to avoid skewed averages.

3. THE 4 BIG DISCOVERIES

1. Digital Boarding is #1

Smooth mobile check-in is the single biggest predictor of happiness. Rating it 4–5/5 produces **82% satisfied flyers**. Rating it ≤3 drops happiness to **13%**.

2. Economy Cabin Crisis

Only **19.4%** of Economy flyers leave satisfied (vs. **69.5%** in Business). Economy makes up over half of our volume and **73%** of all unhappy customers.

3. Comfort Beats Food 2:1

Seat comfort and inflight entertainment correlate twice as strongly with loyalty as food quality. Upgrading catering has almost zero impact on overall passenger ratings.

4. Delays Don't Ruin Loyalty

Even during 60+ minute delays, **65% of flyers stay satisfied** if cabin service is great. But on-time flights with bad service only achieve **30% satisfaction**.

4. THE BIG SURPRISE: THE "CONVENIENT FLIGHT TIMES" MYSTERY

The Anomaly: Why Did Convenient Flights Have Worse Reviews?

At first glance, passengers who rated flight schedules as "convenient" were **less satisfied overall** (negative correlation $r = -0.065$). This sounded completely backwards.

Why did this happen? (Simpson's Paradox):

- **Vacation / Personal Travelers:** Love convenient midday and weekend hours (give high ratings for timing: 3.6/5). But they fly Economy and are generally frustrated with tight seating, paying out-of-pocket, and baggage rules (only **10% satisfied** overall).
- **Business Travelers:** Often fly early morning or late night flights (rate timing lower: 2.8/5). But they fly Business Class with spacious seats and perks, making them **59% satisfied** overall.

Management Takeaway: When you mix both groups together, the large group of unhappy vacationers pulls down the score for convenient flight times. *A great schedule cannot make up for an uncomfortable Economy flight!*

5. WHAT MANAGEMENT SHOULD DO (ACTION PLAN)

We prioritized recommendations based on **fastest impact and lowest cost**:

Recommended Action	Department	Why It Works	How to Measure
1. Rebuild Mobile Check-in Priority 1 (Immediate)	Digital / IT & Airport Teams	One-tap check-in, Apple/Google Wallet boarding passes, and self-scan gate lanes. High impact at low software cost.	App rating $\geq 4.2/5$; mobile adoption $>75\%$.
2. Delay Care Playbook Priority 2 (High ROI)	Customer Experience & Ground Crew	Send automated SMS delay alerts; auto-credit \$10 snack/drink vouchers after 45m delays; unlock free Wi-Fi on tarmac.	Reduced complaint tickets; higher post-delay NPS.
3. Fix Economy Basics Priority 3 (Strategic)	Fleet Maintenance & Procurement	Reallocate catering budget into seat cushions, USB-C power ports, and phone streaming entertainment instead of expensive meal tweaks.	Economy satisfaction target: lift from 19% to 35%+.

6. DATA LIMITATIONS & AI STATEMENT

CAUTION FOR LEADERSHIP

Don't ignore on-time performance: Even though delays don't immediately ruin survey scores, long delays cost millions in crew overtime, gate fines, and missed connection rebookings that surveys don't track.

AI COLLABORATION & VERIFICATION

AI (Gemini/ChatGPT) helped write the Python cleaning code and flagged the flight timing paradox. **Human check:** We corrected an initial AI code bug that treated "0" ratings (no Wi-Fi) as terrible ratings instead of "Not Applicable".

Goal: Lift overall passenger satisfaction from **43.9% to 55%+** within 12 months using smart digital and cabin priorities.